

**Practice question set 2 (please note that this practice set is for quiz 2: module 5 & 6 )**

1. NovaTech Solutions is actively seeking to acquire Cascade Industries. The primary objectives of the acquisition are to create a higher market growth and to take advantage of the unused debt capacity of the target.

Cascade Industries (target)'s standalone value information:

- The FCFF one year from today ( $FCFF_1$ ) will be \$10 million.
- The FCFF will grow at a constant rate of 4% per year perpetually.
- The target's unlevered beta is 1.
- The target is unlevered.

NovaTech Solutions expects the following merger gains once the target is acquired:

- The FCFF of Cascade industries one year from today ( $FCFF_1$ ) will be \$10 million.
- The target's FCFF will grow at a constant rate of 5% per year perpetually.
- The acquirer will maintain a D/E of 1 after the M&A.
- The cost of debt will be 5%.

The risk-free rate is 3%. The market risk premium is 6%. The corporate tax rate is 30%. The target firm has zero non-controlling interests and zero non-total assets. There are no other sources of merger gains other than the ones listed above.

What is the total value of all these merger gains?

*Note there are two ways to calculate the value of synergy:*

*(1)  $V_{AB} - (V_A + V_B)$ , where  $V_{AB}$  is the combined firm value including synergy.  $V_A$  and  $V_B$  are the standalone values of the acquirer and the target, respectively.*

*(2)  $V_B^* - V_B$ , where  $V_B^*$  and  $V_B$  are the target's values with and without synergy, respectively.*

*Since we are given the info of  $V_B^*$ , we will use approach (2).*

Step 1: The Standalone value of the target

- Step 1.1: WACC

- Target's standalone firm value means the target's status quo firm value. Therefore, we should use the target's inputs.
- Target's is unlevered:  $\beta_e = \beta_u = 1$
- Target's cost of equity  $r_e = r_f + \beta_e \times \text{MRP} = 3\% + 1 \times 6\% = 9\%$
- Target's WACC

- $\text{WACC} = r_e = 9\%$

- Step 1.2: Target standalone value

- The FCFFs will grow at a constant rate, so they are a perpetuity.
- $V(\text{total assets}) = \frac{FCFF_1}{r-g} = \frac{10}{0.09-0.04} = \$200 \text{ mln}$

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**Step 2: The value of the target including merger gains**

• Step 2.1: WACC

- There is no change in the target's business nature, so we should use the target's unlevered beta.
- Target's levered beta using the acquirer's D/E :  $\beta_e = \beta_u \times \left[ 1 + \frac{D}{E} \times (1 - T_C) \right] = 1 \times [1 + 1 \times (1 - 0.3)] = 1.7$
- Target's cost of equity  $r_e = r_f + \beta_e \times \text{MRP} = 3\% + 1.7 \times 6\% = 13.2\%$
- Target's WACC using the acquirer's  $r_d$  and D/E
  - $W_d = \frac{D}{D+E} = \frac{1}{1+1} = 0.5$
  - $W_e = \frac{E}{D+E} = \frac{1}{1+1} = 0.5$
  - $\text{WACC} = W_d \times r_d \times (1 - T_C) + W_e \times r_e = 0.5 \times 5\% \times (1 - 0.3) + 0.5 \times 13.2\% = 8.35\%$

• Step 2.2: Target's firm value, including the merger gains

- The FCFFs will grow at a constant rate, so they are a perpetuity.
- Due to the soft synergy (revenue growth): the  $g = 5\%$
- Due to the financial synergy, the WACC will be 8.35%
- $V(\text{total assets}) = \frac{\text{FCFF}_1}{r - g} = \frac{10}{0.0835 - 0.05} = \$298 \text{ mln}$

**Step 3: The value of the merger gains**

- $\text{Total Value of Merger gains} = V(\text{total assets}) - V(\text{total assets}) = \$298 - \$200 = \$98 \text{ mln}$

**2. Your company plans to acquire 100% of Welling Inc.'s equity.**

You are given the following forecasts of the target firm when merger gains are excluded:

- The FCFF one year from today ( $\text{FCFF}_1$ ) will be \$5 million.
- The FCFF will grow at a constant rate of 3% per year perpetually.
- The target's unlevered beta is 0.8.
- The target's cost of debt is 6%.
- The target's current D/E is 0.1. The target company expects to maintain its D/E ratio at D/E at 0.1 in the long run if it is not acquired.

You are also given the following forecasts of the target firm when merger gains are included:

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- The FCFF one year from today ( $FCFF_1$ ) will be \$5.5 million due to some cost saving synergy.
- The FCFF will grow at a constant rate of 3% per year perpetually.
- Due to some financial synergies, the acquirer's cost of debt is 5%.
- The acquirer will maintain a D/E of 1 after the M&A.

The risk-free rate is 3%. The market risk premium is 6%. The corporate tax rate is 30%. The target firm has zero non-controlling interests and zero non-total assets. There are no other sources of merger gains other than the ones listed above.

(a) What is the total value of all these merger gains?

**Step 1: The Standalone value of the target**

- Step 1.1: WACC
  - Target's standalone firm value means the target's status quo firm value. Therefore, we should use the target's inputs.
  - Target's levered beta  $\beta_e = \beta_u \times \left[1 + \frac{D}{E} \times (1 - T_C)\right] = 0.8 \times [1 + 0.1 \times (1 - 0.3)] = 0.856$
  - Target's cost of equity  $r_e = r_f + \beta_e \times MRP = 3\% + 0.856 \times 6\% = 8.136\%$
  - Target's WACC
    - $W_d = \frac{D}{D+E} = \frac{0.1}{0.1+1} = 0.091$
    - $W_e = \frac{E}{D+E} = \frac{1}{0.1+1} = 0.909$
    - $WACC = W_d \times r_d \times (1 - T_C) + W_e \times r_e = 0.091 \times 6\% \times (1 - 0.3) + 0.909 \times 8.136\% = 7.78\%$
- Step 1.2: Target standalone value
  - The FCFFs will grow at a constant rate, so they are a perpetuity.
  - $V(\text{total assets}) = \frac{FCFF_1}{r-g} = \frac{5}{0.078-0.03} = \$104.64 \text{ mln}$

**Step 2: The value of the target including synergies**

- Step 2.1: WACC
  - There is no change in the target's business nature, so we should use the target's unlevered beta.
  - Target's levered beta using the acquirer's D/E :  $\beta_e = \beta_u \times \left[1 + \frac{D}{E} \times (1 - T_C)\right] = 0.8 \times [1 + 1 \times (1 - 0.3)] = 1.36$
  - Target's cost of equity  $r_e = r_f + \beta_e \times MRP = 3\% + 1.3 \times 6\% = 11.16\%$
  - Target's WACC using the acquirer's  $r_d$  and D/E

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- $W_d = \frac{D}{D+E} = \frac{1}{1+1} = 0.5$
- $W_e = \frac{E}{D+E} = \frac{1}{1+1} = 0.5$
- $WACC = W_d \times r_d \times (1 - T_c) + W_e \times r_e = 0.5 \times 5\% \times (1 - 0.3) + 0.5 \times 11.16\% = 7.33\%$
- Step 2.2: Target's firm value, including the synergies
  - The FCFFs will grow at a constant rate, so they are a perpetuity.
  - Due to the hard synergy, the  $FCFF_1$  will be \$5.5
  - Due to the financial synergy, the WACC will be 7.33%
  - $V(\text{total assets}) = \frac{FCFF_1}{r-g} = \frac{5.5}{0.0733-0.03} = \$127.02 \text{ mln}$

Step 3: The value of the these synergies

- $\text{Total Synergy Value} = V(\text{total assets}) - V(\text{total assets}) = \$127.02 - \$104.64 = \$22.38 \text{ mln}$

(b) Assume that the target's company's market value of debt is \$9.52 million. What should be the maximum total offer price?

Target's standalone firm value = \$104.64 mln

So the target's standalone equity value = \$104.64 - \$9.52 = \$95.12 mln

If we assume that the target's pre-M&A market value of equity is the same as the standalone equity value, then the minimum total offer price will be the standalone equity value of the target.

The maximum total offer price will be the standalone equity value of the target + merger gains  
From part (2), the target's standalone equity value = \$104.64 × 0.909 = \$95.12 mln.

From part (3), the value of merger gains = \$22.38 mln.

The maximum offer price = \$95.12 mln + \$22.38 mln = \$117.50 mln.

3. Arctic Cooling Inc. has been in merger talks with Expert HVAC Ltd. Currently both firms are financed with 100% equity. Arctic Cooling, the acquirer, has a market cap of 600 million dollars and 60 million shares outstanding, whereas Expert HVAC has a market cap of 200 million dollars and 50 million shares outstanding.

The management of Arctic Cooling is optimistic toward the synergies between the two companies and estimates the total merger gain would be \$200 million.

The two companies are in the negotiation process and Expert HVAC Ltd tells the acquirer the acceptable total offer price should be \$250 million.

Under this offer price, what is the NPV of the merger deal from the target shareholders' perspective? What is the NPV of the merger deal from the acquirer shareholders' perspective? Is this offer price acceptable to the acquirer?

The total NPV generated by this M&A deal = merger gains = \$200 million.

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The target's piece of NPV = total acquisition premium = total offer price paid – standalone equity of the target = \$250 million - \$200 million = \$50 million.

The acquirer's piece of NPV = \$200 - \$50 = \$150.

Yes, the offer price is acceptable to the acquirer, since it is positive NPV for the acquirer.

4. Middle Crust Resources Ltd. has been in merger talks with GCB group. Currently both firms are financed with 100% equity. Middle Crust Resources, the acquirer, has a market cap of 600 million dollars, whereas GCG Group has a market cap of 200 million dollars. Both firms have 100 million shares outstanding.

The management of the acquirer estimates the total merger gain would be \$100 million.

GCP Group's board wants to negotiate for the highest offer price possible. What will be the highest offer price per share acceptable to Middle Crust Resources?

Assuming that the acquirer will not accept negative NPV M&A deals. Then the highest NPV for the target will be the same as total merger gains.

- The target's NPV = total acquisition premium = all the merger gains = \$100 million
- The acquirer's NPV in this scenario = total merger gains – the target's NPV = \$100 million - \$100 million.

The target's standalone equity value = \$200 million.

Total offer price = the target's standalone equity value + total acquisition premium = \$200 million + \$100 million = \$300 million.

Offer price per share = \$300 million/100 million = \$3 per share.

- This is the maximum offer price acceptable to the acquirer.
- Under this offer price, the acquirer has zero NPV and the target in total has \$100 million of NPV.
- If the offer price is higher than this number, the acquirer will have a negative NPV, and therefore withdraw from negotiation.

5. You are given the following information regarding the acquirer and the target of an M&A deal:

The acquirer:

- WACC = 10%.
- FCFF one year from today ( $FCFF_1$ ) \$2 million. After that, the FCFFs are expected to grow at a constant rate of 3% per year.

The target:

- WACC = 10%.

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- FCFF one year from today ( $FCFF_1$ ) \$1 million. After that, the FCFFs are expected to grow at a constant rate of 2% per year.

Both the acquirer and the target have zero non-controlling interests and zero non-total assets.

The acquirer expects that after the merger, the combined firm will have an  $FCFF_1$  of \$3.1 million and the FCFFs will grow at a constant rate of 3% per year due to some synergies. The combined firm has a WACC of 10%.

Question: What is the value of synergies?

*Note there are two ways to calculate the value of synergy:*

*(1)  $V_{AB} - (V_A + V_B)$ , where  $V_{AB}$  is the combined firm value including synergy.  $V_A$  and  $V_B$  are the standalone values of the acquirer and the target, respectively.*

*(2)  $V_B^* - V_B$ , where  $V_B^*$  and  $V_B$  are the target's values with and without synergy, respectively.*

*Since we are given the info of  $V_A$ ,  $V_B$  and  $V_{AB}$ . we will use approach (a).*

Step 1:  $V_A$

- Note: in order to calculate the acquirer's standalone value  $V_A$ , we should use the acquirer's WACC and cash flows.
- $V_A = FCFF_1 / (WACC - g) = 2 / (0.1 - 0.03) = 28.57$

Step 2:  $V_B$

- Note: in order to calculate the target's standalone value  $V_B$ , we should use the target's WACC and cash flows.
- $V_B = FCFF_1 / (WACC - g) = 1 / (0.1 - 0.02) = 12.5$

Step 3:  $V_{AB}$

- Note: in order to calculate the combined company's standalone value  $V_{AB}$ , we should use the combined company's WACC and cash flows.
- $V_{AB} = FCFF_1 / (WACC - g) = 3.1 / (0.1 - 0.03) = 44.29$

Step 4: Value of synergies

- $V_{AB} - (V_A + V_B) = 44.29 - (28.57 + 12.5) = 3.22$  million

6. You are given the following information regarding the target of an M&A deal:

The target:

- WACC = 10%.
- FCFF one year from today ( $FCFF_1$ ) \$1 million. After that, the FCFFs are expected to grow at a constant rate of 2% per year.

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The target has zero non-controlling interests and zero non-total assets.

The acquirer expects that after the merger, the constant growth rate of the target's FCFFs will be 3% due to cost saving. The WACC and  $FCFF_1$  of the target will remain the same.

What is the value of this hard synergy?

*Note there are two ways to calculate the value of synergy:*

*(1)  $V_{AB} - (V_A + V_B)$ , where  $V_{AB}$  is the combined firm value including synergy.  $V_A$  and  $V_B$  are the standalone values of the acquirer and the target, respectively.*

*(2)  $V_B^* - V_B$ , where  $V_B^*$  and  $V_B$  are the target's values with and without synergy, respectively.*

*Since we are given the info of  $V_B^*$ , we will use approach (2).*

Step 1:  $V_B$

- Note: in order to calculate the target's standalone value  $V_B$ , we should use the target's WACC and cash flows.
- $V_B = FCFF_1 / (WACC - g) = 1 / (0.1 - 0.02) = \$12.5$

Step 2:  $V_B^*$

- $V_B^* = FCFF_1 / (WACC - g) = 1 / (0.1 - 0.03) = \$14.29$

Step 3: Synergy

- Synergy =  $V_B^* - V_B = 14.29 - 12.5 = 1.79$

7. You are given the following information regarding the target of an M&A deal:

The target:

- WACC = 10%.
- FCFF one year from today ( $FCFF_1$ ) \$1 million. After that, the FCFFs are expected to grow at a constant rate of 2% per year.

The target has zero non-controlling interests and zero non-total assets.

The acquirer expects that after the merger, the constant growth rate of the target's FCFFs will be 3% due to some cost saving. This is the only expected synergy. The WACC and  $FCFF_1$  of the target will remain the same.

Assume that the target is unlevered. The acquirer decides to acquire 100% of the target's equity by paying an offer price of \$15 million in cash. What is the NPV of the M&A deal from the acquirer's perspective?

*Similar to the previous question, we will use approach (2) to calculate hard synergy.*

Step 1:

- $V_B = FCFF_1 / (WACC - g) = 1 / (0.1 - 0.02) = \$12.5$

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- $V_B^* = \text{FCCF1} / (\text{WACC} - g) = 1 / (0.1 - 0.03) = \$14.29$
- $\text{Synergy} = V_B^* - V_B = 14.29 - 12.5 = 1.79$

**Step 2:**

- Total NPV (to be shared by acquirer and target) = all the merger gains of this M&A deal = 1.79 since this is the only expected synergy.
- Since the target is unlevered, then its equity value is the same as the value of its total assets (also called total firm value).
- Target's NPV = acquisition premium = total offer price – standalone of the target's equity sold = 15 million – 12.5 million = 2.5 million.
- Acquirer's NPV = total NPV – the target's NPV = 1.79 – 2.5 = -0.71million.

Note that the acquirer has a negative NPV even if there is some synergy. The reason is that the offer price is too high.

8. NovaEdge Corp. intends to acquire 100% of BrightCore Solutions' equity. The standalone equity value of BrightCore Solutions is estimated at \$150 million, and the total expected merger gains are \$50 million.

If the total offer price is \$170 million, what is the NPV of the deal from the acquirer's perspective?

Step 1: NPV from the target's perspective = total acquisition premium

= Total offer price – the target's standalone total equity value = 170 - 150 = \$20 million.

Step 2: NPV from the acquirer's perspective = Total NPV generated by the M&A deal (which is the total merger gains) – the target's NPV from the M&A deal from step 1 = \$50 - \$20 = \$30 million.

9. NovaEdge Corp. intends to acquire 100% of BrightCore Solutions' equity. The standalone equity value of BrightCore Solutions is estimated at \$150 million, and the total expected merger gains are \$50 million.

If the total offer price is \$170 million, what is the NPV of the deal from the target's perspective?

NPV from the target's perspective = total acquisition premium

= Total offer price – the target's standalone total equity value = 170 - 150 = \$20 million.



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10. SkyBridge Technologies plans to acquire 100% of QuantumSoft Systems' equity. The standalone equity value of QuantumSoft Systems is estimated at \$40 million, and the total expected merger gains are \$30 million.

If the acquisition premium is \$50 million, what is the NPV of the deal from the acquirer's perspective?

Step 1: NPV from the target's perspective = total acquisition premium  
= \$50 million.

Step 2: NPV from the acquirer's perspective = Total NPV generated by the M&A deal  
(which is the total merger gains) – the target's NPV from the M&A deal from step 1 = \$30 -  
\$50 = -\$20 million.

11. SkyBridge Technologies plans to acquire 100% of QuantumSoft Systems' equity. The standalone equity value of QuantumSoft Systems is estimated at 40 million, and the total expected merger gains are \$30 million.

If the acquisition premium is \$50 million, what is the NPV of the deal from the target's perspective?

NPV from the target's perspective = total acquisition premium  
= \$50 million.